

```
import pandas as pd
```

```
df = pd.read_csv("students_data.csv")
df
```

	student_id	name	age	gender	grade	math_score	english_score	science_score	enrolled_date	remarks
0	100	Zunaira AHMED	16.0	female	11	75.0	NaN	66	10/06/2022	excellent
1	101	shams abid	16.0	Male	10th	74.0	95	94	10/06/2022	GOOD
2	102	hamza SHAH	NaN	MALE	10	NaN	missing	69	06/12/2022	needs improvement
3	103	Zunaira AHMED	16.0	FEMALE	10	NaN	missing	62	10/06/2022	average
4	104	SARA ALI	16.0	male	11	NaN	96	64	10/06/2022	GOOD
5	105	Maha khan	16.0	Female	10	NaN	NaN	83	06/12/2022	needs improvement
6	106	aliza nasir	17.0	female	11	64.0	NaN	75	06/12/2022	Good
7	107	SARA ALI	17.0	female	12	NaN	63	62	11/06/2022	excellent
8	108	Maha khan	16.0	Female	12	80.0	missing	89	06/12/2022	poor
9	109	nida Zubair	17.0	female	12	NaN	missing	97	10/06/2022	needs improvement
10	110	ahmed noor	17.0	male	11th	65.0	67	100	06/12/2022	excellent
11	111	aliza nasir	16.0	FEMALE	11th	NaN	missing	95	10/06/2022	average
12	112	SAIRA nasir	17.0	female	11	NaN	87	89	10/06/2022	poor
13	113	SAIRA nasir	17.0	female	10	NaN	NaN	98	06/12/2022	Average
14	114	aliza nasir	17.0	male	12	NaN	91	67	11/06/2022	poor
15	115	ahmed noor	16.0	Female	12	65.0	91	94	06/12/2022	Average
16	116	hamza SHAH	NaN	Female	11	NaN	NaN	72	11/06/2022	excellent
17	117	aliza nasir	18.0	male	11th	67.0	74	81	10/06/2022	GOOD
18	118	aliza nasir	17.0	Male	10	100.0	74	62	11/06/2022	average
19	119	SAIRA nasir	17.0	MALE	11th	73.0	NaN	90	06/12/2022	Good
20	120	SARA ALI	17.0	Male	10	NaN	missing	89	10/06/2022	average
21	121	shams abid	18.0	female	11	66.0	72	94	10/06/2022	Average
22	122	SARA ALI	17.0	MALE	11th	75.0	NaN	66	10/06/2022	good student
23	123	Zunaira AHMED	17.0	Female	11th	NaN	missing	63	06/12/2022	excellent
24	124	shams abid	18.0	MALE	11th	NaN	NaN	91	06/12/2022	GOOD
25	125	Maha khan	17.0	Male	12	94.0	80	63	10/06/2022	Average
26	126	shams abid	NaN	Male	10th	64.0	missing	67	06/12/2022	good student
27	127	aliza nasir	16.0	FEMALE	12	NaN	76	80	06/12/2022	poor

```
#df.set_index("student_id")
#df.reset_index()
```

```
df.info
```

```
0      100      Zunaira AHMED      16.0      female      11      75.0      NaN      66      10/06/2022      excellent
7      107      SARA ALI      17.0      female      12      NaN      63
8      108      Maha khan      16.0      Female      12      80.0      missing
9      109      nida Zubair      17.0      female      12      NaN      missing
10     110      ahmed noor      17.0      male      11th      65.0      67
11     111      aliza nasir      16.0      FEMALE      11th      NaN      missing
12     112      SAIRA nasir      17.0      female      11      NaN      87
13     113      SAIRA nasir      17.0      female      10      NaN      NaN
14     114      aliza nasir      17.0      male      12      NaN      91
```

23	123	Zunaira AHMED	17.0	Female	11th	NaN	missing
24	124	shams abid	18.0	MALE	11th	NaN	NaN
25	125	Maha khan	17.0	Male	12	94.0	80
26	126	shams abid	NaN	Male	10th	64.0	missing
27	127	aliza nasir	16.0	FEMALE	12	NaN	76
28	128	SARA ALI	17.0	female	11	NaN	64
29	129	SAIRA nasir	17.0	FEMALE	11	NaN	64
30	129	SAIRA nasir	17.0	FEMALE	11	NaN	64

	science_score	enrolled_date	remarks
0	66	10/06/2022	excellent
1	94	10/06/2022	GOOD
2	69	06/12/2022	needs improvement
3	62	10/06/2022	average
4	64	10/06/2022	GOOD
5	83	06/12/2022	needs improvement
6	75	06/12/2022	Good
7	62	11/06/2022	excellent
8	89	06/12/2022	poor
9	97	10/06/2022	needs improvement
10	100	06/12/2022	excellent
11	95	10/06/2022	average
12	89	10/06/2022	poor
13	98	06/12/2022	Average
14	67	11/06/2022	poor
15	94	06/12/2022	Average
16	72	11/06/2022	excellent
17	81	10/06/2022	GOOD
18	62	11/06/2022	average
19	90	06/12/2022	Good
20	89	10/06/2022	average
21	94	10/06/2022	Average
22	66	10/06/2022	good student
23	63	06/12/2022	excellent
24	91	06/12/2022	GOOD
25	63	10/06/2022	Average
26	67	06/12/2022	good student
27	80	06/12/2022	poor
28	89	10/06/2022	average
29	83	11/06/2022	Average
30	83	11/06/2022	Average >

df.dtypes

```

student_id      int64
name            object
age             float64
gender          object
grade           object
math_score      float64
english_score   object
science_score   int64
enrolled_date   object
remarks         object
dtype: object

```

▼ Data Cleaning

```
df.duplicated() ##mask the duplicate rows { True ==> Duplicate , False ==> Not Duplicate}
```

```

0    False
1    False
2    False
3    False
4    False
5    False
6    False
7    False
8    False
9    False
10   False
11   False
12   False
13   False
14   False
15   False
16   False
17   False
18   False
19   False
20   False
21   False
22   False
23   False
24   False
25   False

```

```
26 False
27 False
28 False
29 False
30 True
dtype: bool
```

```
df = df.drop_duplicates() ##drop duplicates rows
df
```

	student_id	name	age	gender	grade	math_score	english_score	science_score	enrolled_date	remarks
0	100	Zunaira AHMED	16.0	female	11	75.0	NaN	66	10/06/2022	excellent
1	101	shams abid	16.0	Male	10th	74.0	95	94	10/06/2022	GOOD
2	102	hamza SHAH	NaN	MALE	10	NaN	missing	69	06/12/2022	needs improvement
3	103	Zunaira AHMED	16.0	FEMALE	10	NaN	missing	62	10/06/2022	average
4	104	SARA ALI	16.0	male	11	NaN	96	64	10/06/2022	GOOD
5	105	Maha khan	16.0	Female	10	NaN	NaN	83	06/12/2022	needs improvement
6	106	aliza nasir	17.0	female	11	64.0	NaN	75	06/12/2022	Good
7	107	SARA ALI	17.0	female	12	NaN	63	62	11/06/2022	excellent
8	108	Maha khan	16.0	Female	12	80.0	missing	89	06/12/2022	poor
9	109	nida Zubair	17.0	female	12	NaN	missing	97	10/06/2022	needs improvement
10	110	ahmed noor	17.0	male	11th	65.0	67	100	06/12/2022	excellent
11	111	aliza nasir	16.0	FEMALE	11th	NaN	missing	95	10/06/2022	average
12	112	SAIRA nasir	17.0	female	11	NaN	87	89	10/06/2022	poor
13	113	SAIRA nasir	17.0	female	10	NaN	NaN	98	06/12/2022	Average
14	114	aliza nasir	17.0	male	12	NaN	91	67	11/06/2022	poor
15	115	ahmed noor	16.0	Female	12	65.0	91	94	06/12/2022	Average
16	116	hamza SHAH	NaN	Female	11	NaN	NaN	72	11/06/2022	excellent
17	117	aliza nasir	18.0	male	11th	67.0	74	81	10/06/2022	GOOD
18	118	aliza nasir	17.0	Male	10	100.0	74	62	11/06/2022	average
19	119	SAIRA nasir	17.0	MALE	11th	73.0	NaN	90	06/12/2022	Good
20	120	SARA ALI	17.0	Male	10	NaN	missing	89	10/06/2022	average
21	121	shams abid	18.0	female	11	66.0	72	94	10/06/2022	Average
22	122	SARA ALI	17.0	MALE	11th	75.0	NaN	66	10/06/2022	good student
23	123	Zunaira AHMED	17.0	Female	11th	NaN	missing	63	06/12/2022	excellent
24	124	shams abid	18.0	MALE	11th	NaN	NaN	91	06/12/2022	GOOD
25	125	Maha khan	17.0	Male	12	94.0	80	63	10/06/2022	Average
26	126	shams abid	NaN	Male	10th	64.0	missing	67	06/12/2022	good student
27	127	aliza nasir	16.0	FEMALE	12	NaN	76	80	06/12/2022	poor
28	128	SARA ALI	17.0	female	11	NaN	64	89	10/06/2022	average
29	129	SAIRA nasir	17.0	FEMALE	11	NaN	64	83	11/06/2022	Average

```
df.loc[0,"name"]
df.loc["name"] = df["name"].str.lower().str.title() ##converting the names to lowecase and then title case
df.head()
# df["name"] = df["name"].str.lower().str.title()
# df.loc[:, "name"] = df["name"].str.lower().str.title()
# df
```

	student_id	name	age	gender	grade	math_score	english_score	science_score	enrolled_date	remarks
0	100.0	Zunaira Ahmed	16.0	female	11	75.0	NaN	66.0	10/06/2022	excellent
1	101.0	Shams Abid	16.0	Male	10th	74.0	95	94.0	10/06/2022	GOOD
2	102.0	Hamza Shah	NaN	MALE	10	NaN	missing	69.0	06/12/2022	needs improvement
3	103.0	Zunaira Ahmed	16.0	FEMALE	10	NaN	missing	62.0	10/06/2022	average
4	104.0	Sara Ali	16.0	male	11	NaN	96	64.0	10/06/2022	GOOD

```
df["age"].unique() ##display the ages used commonly
```

```
array([16., nan, 17., 18.])
```

```
age_mean = df["age"].mean().round(0) ##take the avg of the age column and round it
age_mean
```

```
np.float64(17.0)
```

```
## fill the mean_age value to the rows having Nan in Age column
df.loc[:, "age"] = df["age"].fillna(age_mean)
df
```

	student_id	name	age	gender	grade	math_score	english_score	science_score	enrolled_date	remarks
0	100.0	Zunaira Ahmed	16.0	female	11	75.0	NaN	66.0	10/06/2022	excellent
1	101.0	Shams Abid	16.0	Male	10th	74.0	95	94.0	10/06/2022	GOOD
2	102.0	Hamza Shah	17.0	MALE	10	NaN	missing	69.0	06/12/2022	needs improvement
3	103.0	Zunaira Ahmed	16.0	FEMALE	10	NaN	missing	62.0	10/06/2022	average
4	104.0	Sara Ali	16.0	male	11	NaN	96	64.0	10/06/2022	GOOD
5	105.0	Maha Khan	16.0	Female	10	NaN	NaN	83.0	06/12/2022	needs improvement
6	106.0	Aliza Nasir	17.0	female	11	64.0	NaN	75.0	06/12/2022	Good
7	107.0	Sara Ali	17.0	female	12	NaN	63	62.0	11/06/2022	excellent
8	108.0	Maha Khan	16.0	Female	12	80.0	missing	89.0	06/12/2022	poor
9	109.0	Nida Zubair	17.0	female	12	NaN	missing	97.0	10/06/2022	needs improvement
10	110.0	Ahmed Noor	17.0	male	11th	65.0	67	100.0	06/12/2022	excellent
11	111.0	Aliza Nasir	16.0	FEMALE	11th	NaN	missing	95.0	10/06/2022	average
12	112.0	Saira Nasir	17.0	female	11	NaN	87	89.0	10/06/2022	poor
13	113.0	Saira Nasir	17.0	female	10	NaN	NaN	98.0	06/12/2022	Average
14	114.0	Aliza Nasir	17.0	male	12	NaN	91	67.0	11/06/2022	poor
15	115.0	Ahmed Noor	16.0	Female	12	65.0	91	94.0	06/12/2022	Average
16	116.0	Hamza Shah	17.0	Female	11	NaN	NaN	72.0	11/06/2022	excellent
17	117.0	Aliza Nasir	18.0	male	11th	67.0	74	81.0	10/06/2022	GOOD
18	118.0	Aliza Nasir	17.0	Male	10	100.0	74	62.0	11/06/2022	average
19	119.0	Saira Nasir	17.0	MALE	11th	73.0	NaN	90.0	06/12/2022	Good
20	120.0	Sara Ali	17.0	Male	10	NaN	missing	89.0	10/06/2022	average
21	121.0	Shams Abid	18.0	female	11	66.0	72	94.0	10/06/2022	Average
22	122.0	Sara Ali	17.0	MALE	11th	75.0	NaN	66.0	10/06/2022	good student
23	123.0	Zunaira Ahmed	17.0	Female	11th	NaN	missing	63.0	06/12/2022	excellent
24	124.0	Shams Abid	18.0	MALE	11th	NaN	NaN	91.0	06/12/2022	GOOD
25	125.0	Maha Khan	17.0	Male	12	94.0	80	63.0	10/06/2022	Average
26	126.0	Shams Abid	17.0	Male	10th	64.0	missing	67.0	06/12/2022	good student
27	127.0	Aliza Nasir	16.0	FEMALE	12	NaN	76	80.0	06/12/2022	poor

```
df
```

	student_id	name	age	gender	grade	math_score	english_score	science_score	enrolled_date	remarks
0	100	Zunaira Ahmed	16.0	female	11	75.0	NaN	66	10/06/2022	excellent
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15	115	Ahmed Noor	16.0	Female	12	65.0	91	94	06/12/2022	Average
16	116	Hamza Shah	17.0	Female	11	NaN	NaN	72	11/06/2022	excellent
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23	123	Zunaira Ahmed	17.0	Female	11th	NaN	missing	63	06/12/2022	excellent
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27	127	Aliza Nasir	16.0	FEMALE	12	NaN	76	80	06/12/2022	poor
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29	129	Saira Nasir	17.0	FEMALE	11	NaN	64	83	11/06/2022	Average

```
df["gender"].unique() ##print the common values used in gender column
```

```
array(['female', 'Male', 'MALE', 'FEMALE', 'male', 'Female', nan],
      dtype=object)
```

```
df.loc[:, "gender"] = df["gender"].str.lower().str.title()
```

```
df
```

	student_id	name	age	gender	grade	math_score	english_score	science_score	enrolled_date	remarks
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29	129	Saira Nasir	17.0	Female	11	NaN	64	83	11/06/2022	Average

df["grade"]

0 11
1 10th
2 10
3 10
4 11
5 10
6 11
7 12
8 12
9 12
10 11th
11 11th
12 11
13 10
14 12
15 12
16 11
17 11th
18 10
19 11th
20 10
21 11
22 11th
23 11th
24 11th
25 12
26 10th
27 12
28 11

```
29      11
Name: grade, dtype: object
```

```
df["grade"].unique()
```

```
array(['11', '10th', '10', '12', '11th'], dtype=object)
```

```
df.loc[:, "grade"] = df["grade"].replace({'10th': '10', '11th': '11'})
```

```
df
```

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16	116	Hamza Shah	17.0	Female	11	NaN	NaN	72	11/06/2022	excellent
17	117	Aliza Nasir	18.0	Male	11	67.0	74	81	10/06/2022	GOOD
18	118	Aliza Nasir	17.0	Male	10	100.0	74	62	11/06/2022	average
19	119	Saira Nasir	17.0	Male	11	73.0	NaN	90	06/12/2022	Good
20	120	Sara Ali	17.0	Male	10	NaN	missing	89	10/06/2022	average
21	121	Shams Abid	18.0	Female	11	66.0	72	94	10/06/2022	Average
22	122	Sara Ali	17.0	Male	11	75.0	NaN	66	10/06/2022	good student
23	123	Zunaira Ahmed	17.0	Female	11	NaN	missing	63	06/12/2022	excellent
24	124	Shams Abid	18.0	Male	11	NaN	NaN	91	06/12/2022	GOOD
25	125	Maha Khan	17.0	Male	12	94.0	80	63	10/06/2022	Average
26	126	Shams Abid	17.0	Male	10	64.0	missing	67	06/12/2022	good student
27	127	Aliza Nasir	16.0	Female	12	NaN	76	80	06/12/2022	poor
28	128	Sara Ali	17.0	Female	11	NaN	64	89	10/06/2022	average
29	129	Saira Nasir	17.0	Female	11	NaN	64	83	11/06/2022	Average

▼ Handle Missing Values

```
df.isnull()
```

	student_id	name	age	gender	grade	math_score	english_score	science_score	enrolled_date	remarks
0	False	False	False	False	False	False	True	False	False	False
1	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	True	False	False	False	False
3	False	False	False	False	False	True	False	False	False	False
4	False	False	False	False	False	True	False	False	False	False
5	False	False	False	False	False	True	True	False	False	False
6	False	False	False	False	False	False	True	False	False	False
7	False	False	False	False	False	True	False	False	False	False
8	False	False	False	False	False	False	False	False	False	False
9	False	False	False	False	False	True	False	False	False	False
10	False	False	False	False	False	False	False	False	False	False
11	False	False	False	False	False	True	False	False	False	False
12	False	False	False	False	False	True	False	False	False	False
13	False	False	False	False	False	True	True	False	False	False
14	False	False	False	False	False	True	False	False	False	False
15	False	False	False	False	False	False	False	False	False	False
16	False	False	False	False	False	True	True	False	False	False
17	False	False	False	False	False	False	False	False	False	False
18	False	False	False	False	False	False	False	False	False	False
19	False	False	False	False	False	False	True	False	False	False
20	False	False	False	False	False	True	False	False	False	False
21	False	False	False	False	False	False	False	False	False	False
22	False	False	False	False	False	False	True	False	False	False
23	False	False	False	False	False	True	False	False	False	False
24	False	False	False	False	False	True	True	False	False	False
25	False	False	False	False	False	False	False	False	False	False
26	False	False	False	False	False	False	False	False	False	False
27	False	False	False	False	False	True	False	False	False	False
28	False	False	False	False	False	True	False	False	False	False
29	False	False	False	False	False	True	False	False	False	False

df.isnull().sum()

```
student_id      0
name            0
age            0
gender         0
grade          0
math_score     17
english_score   8
science_score   0
enrolled_date  0
remarks        0
dtype: int64
```

```
# Select the rows that have at least one missing value
df[df.isnull().any(axis=1)].head()
```

	student_id	name	age	gender	grade	math_score	english_score	science_score	enrolled_date	remarks
0	100	Zunaira Ahmed	16.0	Female	11	75.0	NaN	66	10/06/2022	excellent
2	102	Hamza Shah	17.0	Male	10	NaN	missing	69	06/12/2022	needs improvement
3	103	Zunaira Ahmed	16.0	Female	10	NaN	missing	62	10/06/2022	average
4	104	Sara Ali	16.0	Male	11	NaN	96	64	10/06/2022	GOOD
5	105	Maha Khan	16.0	Female	10	NaN	NaN	83	06/12/2022	needs improvement


```
math_mean = df["math_score"].mean()
```

```
df.loc[:, "math_score"] = df["math_score"].fillna(math_mean)
```

df

	student_id	name	age	gender	grade	math_score	english_score	science_score	enrolled_date	remarks
0	100	Zunaira Ahmed	16.0	Female	11	75.0	NaN	66	10/06/2022	excellent
1	101	Shams Abid	16.0	Male	10	74.0	95.0	94	10/06/2022	GOOD
2	102	Hamza Shah	17.0	Male	10	74.0	NaN	69	06/12/2022	needs improvement
3	103	Zunaira Ahmed	16.0	Female	10	74.0	NaN	62	10/06/2022	average
4	104	Sara Ali	16.0	Male	11	74.0	96.0	64	10/06/2022	GOOD
5	105	Maha Khan	16.0	Female	10	74.0	NaN	83	06/12/2022	needs improvement
6	106	Aliza Nasir	17.0	Female	11	64.0	NaN	75	06/12/2022	Good
7	107	Sara Ali	17.0	Female	12	74.0	63.0	62	11/06/2022	excellent
8	108	Maha Khan	16.0	Female	12	80.0	NaN	89	06/12/2022	poor
9	109	Nida Zubair	17.0	Female	12	74.0	NaN	97	10/06/2022	needs improvement
10	110	Ahmed Noor	17.0	Male	11	65.0	67.0	100	06/12/2022	excellent
11	111	Aliza Nasir	16.0	Female	11	74.0	NaN	95	10/06/2022	average
12	112	Saira Nasir	17.0	Female	11	74.0	87.0	89	10/06/2022	poor
13	113	Saira Nasir	17.0	Female	10	74.0	NaN	98	06/12/2022	Average
14	114	Aliza Nasir	17.0	Male	12	74.0	91.0	67	11/06/2022	poor
15	115	Ahmed Noor	16.0	Female	12	65.0	91.0	94	06/12/2022	Average
16	116	Hamza Shah	17.0	Female	11	74.0	NaN	72	11/06/2022	excellent
17	117	Aliza Nasir	18.0	Male	11	67.0	74.0	81	10/06/2022	GOOD
18	118	Aliza Nasir	17.0	Male	10	100.0	74.0	62	11/06/2022	average
19	119	Saira Nasir	17.0	Male	11	73.0	NaN	90	06/12/2022	Good
20	120	Sara Ali	17.0	Male	10	74.0	NaN	89	10/06/2022	average
21	121	Shams Abid	18.0	Female	11	66.0	72.0	94	10/06/2022	Average
22	122	Sara Ali	17.0	Male	11	75.0	NaN	66	10/06/2022	good student
23	123	Zunaira Ahmed	17.0	Female	11	74.0	NaN	63	06/12/2022	excellent
24	124	Shams Abid	18.0	Male	11	74.0	NaN	91	06/12/2022	GOOD
25	125	Maha Khan	17.0	Male	12	94.0	80.0	63	10/06/2022	Average
26	126	Shams Abid	17.0	Male	10	64.0	NaN	67	06/12/2022	good student
27	127	Aliza Nasir	16.0	Female	12	74.0	76.0	80	06/12/2022	poor
28	128	Sara Ali	17.0	Female	11	74.0	64.0	89	10/06/2022	average
29	129	Saira Nasir	17.0	Female	11	74.0	64.0	83	11/06/2022	Average

df["english_score"]

```
0      NaN
1       95
2    missing
3    missing
4       96
5      NaN
6      NaN
7       63
8    missing
9    missing
10      67
11    missing
12      87
13      NaN
14      91
15      91
16      NaN
17      74
18      74
19      NaN
20    missing
21      72
```

```
22      NaN
23    missing
24      NaN
25       80
26    missing
27       76
28       64
29       64
Name: english_score, dtype: object
```

```
df.loc[:, "english_score"] = pd.to_numeric(df["english_score"], errors='coerce')
```

df

	student_id	name	age	gender	grade	math_score	english_score	science_score	enrolled_date	remarks
0	100	Zunaira Ahmed	16.0	Female	11	75.0	NaN	66	10/06/2022	excellent
1	101	Shams Abid	16.0	Male	10	74.0	95.0	94	10/06/2022	GOOD
2	102	Hamza Shah	17.0	Male	10	74.0	NaN	69	06/12/2022	needs improvement
3	103	Zunaira Ahmed	16.0	Female	10	74.0	NaN	62	10/06/2022	average
4	104	Sara Ali	16.0	Male	11	74.0	96.0	64	10/06/2022	GOOD
5	105	Maha Khan	16.0	Female	10	74.0	NaN	83	06/12/2022	needs improvement
6	106	Aliza Nasir	17.0	Female	11	64.0	NaN	75	06/12/2022	Good
7	107	Sara Ali	17.0	Female	12	74.0	63.0	62	11/06/2022	excellent
8	108	Maha Khan	16.0	Female	12	80.0	NaN	89	06/12/2022	poor
9	109	Nida Zubair	17.0	Female	12	74.0	NaN	97	10/06/2022	needs improvement
10	110	Ahmed Noor	17.0	Male	11	65.0	67.0	100	06/12/2022	excellent
11	111	Aliza Nasir	16.0	Female	11	74.0	NaN	95	10/06/2022	average
12	112	Saira Nasir	17.0	Female	11	74.0	87.0	89	10/06/2022	poor
13	113	Saira Nasir	17.0	Female	10	74.0	NaN	98	06/12/2022	Average
14	114	Aliza Nasir	17.0	Male	12	74.0	91.0	67	11/06/2022	poor
15	115	Ahmed Noor	16.0	Female	12	65.0	91.0	94	06/12/2022	Average
16	116	Hamza Shah	17.0	Female	11	74.0	NaN	72	11/06/2022	excellent
17	117	Aliza Nasir	18.0	Male	11	67.0	74.0	81	10/06/2022	GOOD
18	118	Aliza Nasir	17.0	Male	10	100.0	74.0	62	11/06/2022	average
19	119	Saira Nasir	17.0	Male	11	73.0	NaN	90	06/12/2022	Good
20	120	Sara Ali	17.0	Male	10	74.0	NaN	89	10/06/2022	average
21	121	Shams Abid	18.0	Female	11	66.0	72.0	94	10/06/2022	Average
22	122	Sara Ali	17.0	Male	11	75.0	NaN	66	10/06/2022	good student
23	123	Zunaira Ahmed	17.0	Female	11	74.0	NaN	63	06/12/2022	excellent
24	124	Shams Abid	18.0	Male	11	74.0	NaN	91	06/12/2022	GOOD
25	125	Maha Khan	17.0	Male	12	94.0	80.0	63	10/06/2022	Average
26	126	Shams Abid	17.0	Male	10	64.0	NaN	67	06/12/2022	good student
27	127	Aliza Nasir	16.0	Female	12	74.0	76.0	80	06/12/2022	poor
28	128	Sara Ali	17.0	Female	11	74.0	64.0	89	10/06/2022	average
29	129	Saira Nasir	17.0	Female	11	74.0	64.0	83	11/06/2022	Average

Start coding or [generate](#) with AI.

```
df["remarks"].unique()

array(['excellent', 'GOOD', 'needs improvement', 'average', 'Good',
      'poor', 'Average', 'good student'], dtype=object)
```

TASK: write code to standardize 'Remarks' column

df.loc["remarks"] = df["remarks"].str.lower().str.title()

df

	student_id	name	age	gender	grade	math_score	english_score	science_score	enrolled_date	remarks
0	100.0	Zunaira Ahmed	16.0	Female	11	75.0	NaN	66.0	10/06/2022	Excellent
1	101.0	Shams Abid	16.0	Male	10	74.0	95.0	94.0	10/06/2022	Good
2	102.0	Hamza Shah	17.0	Male	10	74.0	NaN	69.0	06/12/2022	Average
3	103.0	Zunaira Ahmed	16.0	Female	10	74.0	NaN	62.0	10/06/2022	Average
4	104.0	Sara Ali	16.0	Male	11	74.0	96.0	64.0	10/06/2022	Good
5	105.0	Maha Khan	16.0	Female	10	74.0	NaN	83.0	06/12/2022	Average
6	106.0	Aliza Nasir	17.0	Female	11	64.0	NaN	75.0	06/12/2022	Good
7	107.0	Sara Ali	17.0	Female	12	74.0	63.0	62.0	11/06/2022	Excellent
8	108.0	Maha Khan	16.0	Female	12	80.0	NaN	89.0	06/12/2022	Poor
9	109.0	Nida Zubair	17.0	Female	12	74.0	NaN	97.0	10/06/2022	Average
10	110.0	Ahmed Noor	17.0	Male	11	65.0	67.0	100.0	06/12/2022	Excellent
11	111.0	Aliza Nasir	16.0	Female	11	74.0	NaN	95.0	10/06/2022	Average
12	112.0	Saira Nasir	17.0	Female	11	74.0	87.0	89.0	10/06/2022	Poor
13	113.0	Saira Nasir	17.0	Female	10	74.0	NaN	98.0	06/12/2022	Average
14	114.0	Aliza Nasir	17.0	Male	12	74.0	91.0	67.0	11/06/2022	Poor
15	115.0	Ahmed Noor	16.0	Female	12	65.0	91.0	94.0	06/12/2022	Average
16	116.0	Hamza Shah	17.0	Female	11	74.0	NaN	72.0	11/06/2022	Excellent